

Sd Maximum Gas

Symptoms

If the unit is running in dual fuel mode, the system will shut down the flow of gas and all dual fuel operations will cease until the fault is corrected.

How To Use This Tree

This tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Circuit Description

The system monitors diesel fuel flow and compares to a mapped table within the controller. If the gas system is on and there is a difference between the mapped table and actual diesel fuel flow, a ratio of gas to diesel is calculated based on the difference in reported fuel rate and the convert diesel table. This ratio is monitored to make sure the system does **not** supply more gas/diesel substitution to the engine than Cummins Inc. has defined.

Conditions for Running the Diagnostics

Controller is powered ON, system is in 'Auto' Mode, or system is in 'Manual' Mode with a start command. Engine is running with loads and conditions suitable to begin gas flow and the 'GAS OK' lamp is ON.

Conditions for Activating the Fault Message:

The ratio of gas to diesel has exceeded the variable set-point "MaxGasProt" for greater than a fixed value of 3 seconds. Set-point is factory default at 75 percent and password level 2 protected.

Conditions for Clearing the Fault Codes Automatically:

Clearing the fault codes has to be manually reset.

Conditions for Clearing the Fault Codes Manually:

Press "Fault Reset" button on front of controller.

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Possible causes include:

- Throttle body malfunction
- Throttle body feedback malfunction
- Low fuel rate

Service Tip: From the controller display the gas/diesel ratio can be seen from the main screen.

Service Tip: Using the software package “Drive Monitor” and using the “Control Window” the same observations can be made.

Fault Message Sd Maximum Gas		
Gas/Diesel Ratio Has Exceed Set-point Limit and Measures to Reduce the G/D Ratio are Not Reducing the Substitution Level		
Codes or Messages	Reason	Effect
Sd Maximum Gas	Displaced diesel flow and calculated G/D ratio has exceeded limit set- point (80 percent) for 3.0 seconds.	Dual fuel control system will not allow gas operations. Dual fuel controller will stop gas flow.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Validate the fault message.	
	STEP 1A. Check for active fault message.	Sd Maximum Gas alarm code is active?
	STEP 1B. Check for active fault message.	Alarm Codes other than Sd Maximum Gas active?
	STEP 1C. Check the engine ECM for engine fault codes.	Engine ECM indicates fault codes?
STEP 2.	Monitor Dual Fuel operation.	
	STEP 2A. Observe and record operating characteristics.	Did the valve feedback follow the command?
STEP 3.	Check throttle body circuit.	
	STEP 3A. Check throttle body supply.	Voltage between 20-30 VDC?
	STEP 3B. Check throttle body supply.	Voltage between 20-30 VDC?
	STEP 3C. Check throttle body supply circuit breaker 2.	Circuit breaker tripped?
	STEP 3D. Check throttle body supply.	Voltage between 20-30 VDC?
STEP 4.	Check throttle body feedback circuit.	
	STEP 4A. Check throttle body feedback circuit.	Voltage measures 5 VDC?

STEPS		SPECIFICATIONS
	STEP 4B. Check throttle body supply.	Voltage measures 5 VDC?
	STEP 4C. Check reference voltage at DF module.	Voltage measures 5 VDC?
STEP 5.	Check throttle position signal circuit.	
	STEP 5A. Check throttle position signal for open circuit.	Resistance greater than 10 ohms?
	STEP 5B. Check throttle body feedback circuit for pin to pin short.	Resistance less than 100k ohms?

STEP 1. Validate the fault message.

STEP 1A. Check for active fault message.

Conditions :

- Power ON dual fuel control module.
- Connect IntelliMonitor to the dual fuel control panel.

Action	Specification/Repair	Next Step
• Check local display panel or IntelliMonitor for fault messages in alarm list and history.	Sd Maximum Gas alarm code is active? YES	1B
	Sd Maximum Gas alarm code is active? NO	Return the pump to service and monitor.

STEP 1B. Check for active fault message.

Conditions :

- Power ON dual fuel control module.
- Connect IntelliMonitor to the dual fuel control panel.

Action	Specification/Repair	Next Step
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Check local display panel or IntelliMonitor for fault messages in alarm list and history.	Alarm Codes other than Sd Maximum Gas active? YES	Troubleshoot all other error codes prior to troubleshooting Sd Maximum Gas.
	Alarm Codes other than Sd Maximum Gas active? NO	1C

STEP 1C. Check the engine ECM for engine fault codes.

Conditions :

- Turn keyswitch ON.
- Connect the recommended Cummins® electronic service tool or equivalent.

Action	Specification/Repair	Next Step
Use the recommended Cummins® electronic service tool or equivalent to read the fault code.	Engine ECM indicates fault codes? YES	Troubleshoot engine fault codes prior to Troubleshooting Alarm Codes.
	Engine ECM indicates fault codes? NO	2A

STEP 2. Monitor Dual Fuel operation.

STEP 2A. Observe and record operating characteristics.

Conditions :

- Power ON dual fuel controller.
- Verify Dual fuel controller mode is OFF or MANUAL.
- Operate engine above 400 kw.
- Connect IntelliMonitor to the dual fuel control panel.

Action	Specification/Repair	Next Step
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Observe and record the following information from values, gas regulation in InteliMonitor. • EngPwr • MapDiesel • GasVlvCommand • TB Feedback Place controller in 'Auto', observe the dual fuel start up sequence and allow stabilization of Load and G/D ratio. This test must be performed twice before moving onto the next step. In some cases at lower loads and speeds where the fuel rate is pushed down near the Min Dsl Rate (78 l/hr) the fuelrate can become unstable and even drop sharply which will cause the G/D to then increase sharply.	Did the valve feedback follow the command? YES Repair : An error in the fueling table has been identified. Contact a local Cummins® Distributor to recommission the unit.	Repair complete
	Did the valve feedback follow the command? NO	3A

STEP 3. Check throttle body circuit.

STEP 3A. Check throttle body supply.

Conditions :

- Power ON dual fuel control module.
- Disconnect 23 pin connector from Woodward throttle body.

Action	Specification/Repair	Next Step
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Check for battery voltage between supply pins of 23 pin Woodward throttle body connector and return pin in same connector. Reference circuit diagrams for proper pin identification.	Voltage between 20-30 VDC? YES	4A
	Voltage between 20-30 VDC? NO	3B

STEP 3B. Check throttle body supply.

Conditions :

- Power ON dual fuel control module.
- Disconnect C8 panel connector from dual fuel panel.

Action	Specification/Repair	Next Step
Check for battery voltage between throttle body supply pin of C8 connector and return pin of C8 connector. Reference circuit diagrams for proper pin identification.	Voltage between 20-30 VDC? YES Repair : Repair or replace the dual fuel wiring harness.	Repair complete
	Voltage between 20-30 VDC? NO	3C

STEP 3C. Check throttle body supply circuit breaker 2.

Conditions :

- Power ON dual fuel control module.

Action	Specification/Repair	Next Step
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Verify throttle body circuit breaker 2 is not tripped.	Circuit breaker tripped? YES Repair : Reset or replace the circuit breaker.	Repair complete
	Circuit breaker tripped? NO	3D

STEP 3D. Check throttle body supply.

Conditions :

- Disconnect C2 panel connector from dual fuel panel.

Action	Specification/Repair	Next Step
Check for battery voltage between throttle body supply pins of C2 connector and return pins of C2 connector. Reference circuit diagrams for proper pin identification.	Voltage between 20-30 VDC? YES Repair : Repair or replace the dual fuel panel internal wiring harness.	Repair complete
	Voltage between 20-30 VDC? NO Repair : Repair or replace the dual fuel panel external power supply wiring harness.	Repair complete

STEP 4. Check throttle body feedback circuit.

STEP 4A. Check throttle body feedback circuit.

Conditions :

- Power ON dual fuel control module.
- Disconnect 23 pin connector from Woodward throttle body.

Action	Specification/Repair	Next Step
Check for 5V supply on analog signal and analog return in same connector. Reference circuit diagrams for proper pin identification.	Voltage measures 5 VDC? YES	5A
	Voltage measures 5 VDC? NO	4B

STEP 4B. Check throttle body supply.**Conditions :**

- Power ON dual fuel control module.
- Disconnect C8 panel connector from dual fuel panel.

Action	Specification/Repair	Next Step
Check for 5V supply on analog signal and analog return in same connector. Reference circuit diagrams for proper pin identification.	Voltage measures 5 VDC? YES Repair : Repair or replace the dual fuel wiring harness.	Repair complete
	Voltage measures 5 VDC? NO	4C

STEP 4C. Check reference voltage at DF module**Conditions :**

- Power ON dual fuel control module.

Action	Specification/Repair	Next Step
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Measure 5V reference voltage at BF1 connectors of DF Module.	Voltage measures 5 VDC? YES Repair : Repair or replace the dual fuel panel internal wiring harness.	Repair complete
	Voltage measures 5 VDC? NO Repair : Replace dual fuel module.	Repair complete

STEP 5. Check throttle position signal circuit.

STEP 5A. Check throttle position signal for open circuit.

Conditions :

- Power OFF dual fuel control module.
- Disconnect 23 pin connector from Woodward throttle body.

Action	Specification/Repair	Next Step
Measure resistance between throttle position sensor signal pin of 23 pin Woodward throttle body connector and throttle position sensor pin of IBF Module. Reference circuit diagrams for proper pin identification.	Resistance greater than 10 ohms? YES Repair : Repair or replace dual fuel wiring harness.	Repair complete
	Resistance greater than 10 ohms? NO	5B

STEP 5B. Check throttle body feedback circuit for pin to pin short.

Conditions :

- Disconnect 23 pin connector from Woodward throttle body.
- Disconnect feedback circuit pins from IBF module.

Action	Specification/Repair	Next Step
Measure resistance between throttle position sensor signal pin of 23 pin Woodward throttle body connector and all other pins in that connector. Reference circuit diagrams for proper pin identification.	Resistance less than 100k ohms? YES Repair : Repair or replace dual fuel wiring harness.	Repair complete
	Resistance less than 100k ohms? NO Repair : Replace Woodward throttle body. Refer to Procedure 005-052 in Section 5. (/qs3/pubsys2/xml/en/procedures/330/330-005-052.html)	Repair complete